

CS 663: Digital Image Processing

Assignment 1: Problem 1(e) Report

Pushpendra Uikey

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1 Image Subsampling and Interpolation

1.1 1(e) Image Rotation using Bilinear and Nearest-Neighbor Interpolation

Implementation

The objective was to rotate the entire input image (`main.png`) such that the slanted lamp post in the foreground becomes perfectly vertical, while keeping the output image size identical to the input. An experimentally determined counter-clockwise rotation angle of 5.8° was applied. To prevent the formation of empty pixels or “holes” in the output, an inverse mapping approach was utilized. The algorithm iterates over every pixel in the destination grid and maps it back to a floating-point coordinate in the source image using an inverse 2D rotation matrix centered on the image midpoint.

Two distinct interpolation functions were implemented to assign intensity values to these mapped coordinates:

- **Nearest-Neighbor** (`myImageRotationUsingNearestNeighborInterp`): Applies standard rounding to the mapped floating-point coordinates to select the exact intensity of the closest integer pixel.
- **Bilinear** (`myImageRotationUsingBilinearInterp`): Extracts the fractional spatial distances and computes a weighted average from the surrounding 2×2 pixel neighborhood.

In both cases, if the inverse-mapped coordinate falls outside the original image dimensions, the pixel is safely padded with an intensity of 0.0 (black).

Observations and Analysis

The differences between the two interpolation methods are starkly visible along high-contrast diagonal and vertical edges.

- **Nearest-Neighbor Output:** This approach produces prominent aliasing artifacts. The vertical lines of the building’s facade and the targeted lamp post exhibit jagged, stair-stepped boundaries.
- **Bilinear Output:** This method successfully smooths the jagged edges, yielding a much more continuous and natural-looking lamp post. However, this distance-weighted averaging inherently acts as a low-pass filter, slightly softening the overall focus and finer textures in the foliage compared to the nearest-neighbor result.

Additionally, because the output image dimensions are restricted to match the original input, the 5.8° rotation forces the corners of the original image out of frame. The zero-padding logic successfully fills the resulting empty regions at the borders with solid black triangles, preventing edge-smearing.

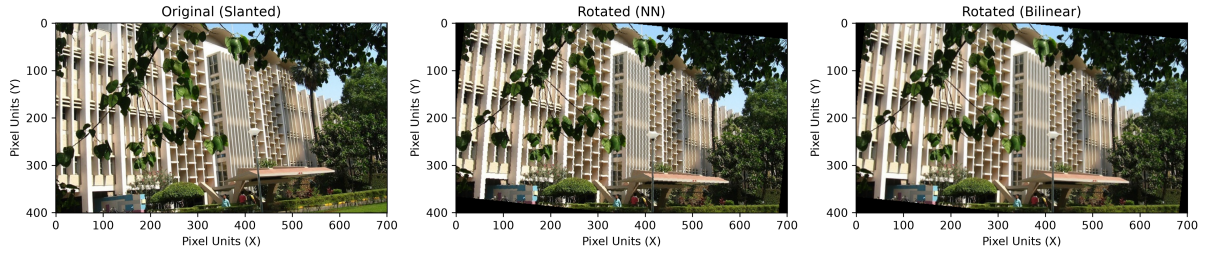


Figure 1: Side-by-side comparison of the original slanted image, the nearest-neighbor rotation, and the bilinear rotation.

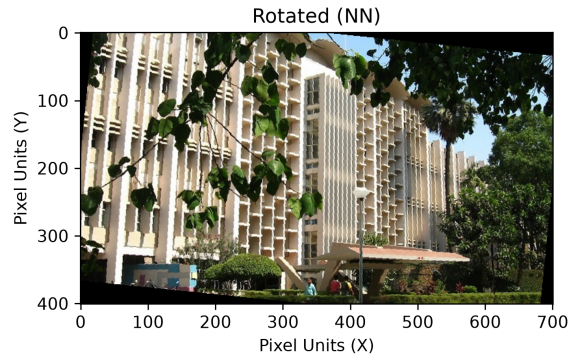


Figure 2: Nearest-Neighbor Rotation. Note the stair-stepped jagged edges on the building and lamp post.

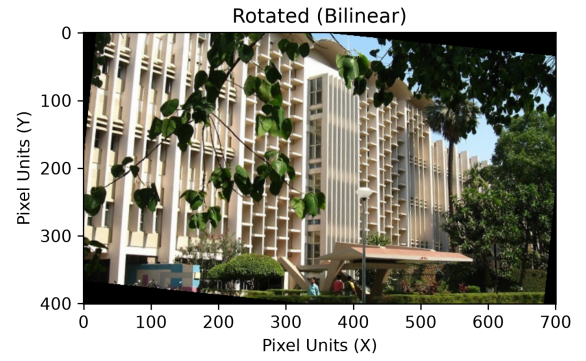


Figure 3: Bilinear Rotation. Edges are significantly smoother, though slightly softened overall.